Regime shift - Macro Aware Tactical Asset Allocation Engine

Executive Summary

Financial markets exhibit non-stationary behavior characterized by structural shifts between expansionary, defensive, and crisis environments. Traditional static portfolio allocation strategies, such as Buy-and-Hold or the classical 60/40 portfolio, implicitly assume stable return distributions and constant asset relationships. In practice, these assumptions frequently fail during periods of elevated uncertainty, leading to significant drawdowns and deteriorating risk-adjusted performance.

This project develops a complete quantitative investment framework that combines probabilistic regime detection with dynamic portfolio optimization. Hidden Markov Models (HMMs) are employed to identify latent market regimes directly from historical market data without requiring manually labeled economic states. The inferred regimes are then used to dynamically adjust portfolio allocations across equities (SPY), long-term Treasury bonds (TLT), and gold (GLD), allowing the strategy to adapt as market conditions evolve.

To ensure methodological rigor, all investment decisions are evaluated using a strict walk-forward backtesting framework. Regime predictions are shifted forward by one trading day to eliminate look-ahead bias, while portfolio turnover is explicitly penalized through transaction cost modeling.

Across approximately twenty years of historical data (2005–2026), the proposed regime-aware strategy consistently outperformed multiple benchmark portfolios, including Buy-and-Hold SPY, Equal Weight, and the traditional 60/40 allocation. The strategy achieved higher cumulative returns, superior Sharpe and Sortino ratios, substantially lower maximum drawdown, and improved downside protection during periods of market stress.

Comprehensive regime analysis further demonstrated that the Hidden Markov Model identified economically meaningful and persistent market states characterized by realistic transition probabilities, stable regime durations, and intuitive market behavior.

Overall, the results suggest that combining probabilistic market regime detection with adaptive portfolio optimization provides a robust and interpretable framework for dynamic asset allocation in non-stationary financial markets.

Problem Statement

Classical portfolio optimization techniques assume that expected returns, volatilities, and correlations remain relatively stable through time. Financial markets, however, are inherently non-stationary. Market dynamics evolve continuously as economies transition between expansion, contraction, recovery, and crisis.

Static allocation strategies are therefore unable to adapt to structural changes in market behavior. During periods of elevated uncertainty, equity-heavy portfolios frequently experience significant drawdowns, while defensive assets become increasingly attractive.

The central objective of this research is to determine whether latent market regimes can be identified directly from financial time series using Hidden Markov Models and whether those regimes can be used to improve portfolio allocation decisions.

Rather than attempting to predict future prices directly, the proposed framework first estimates the current hidden economic environment and then allocates capital according to the statistical characteristics of that environment.

Research Questions

This study investigates the following research questions:

1. Can Hidden Markov Models identify economically meaningful latent market regimes without manual labeling?
2. Do different hidden regimes exhibit statistically distinct return and volatility characteristics?

3. Can portfolio optimization conditioned on hidden market regimes outperform static allocation strategies?
4. Does a walk-forward implementation maintain superior performance after eliminating look-ahead bias?
5. Can adaptive asset allocation improve downside protection while maintaining competitive long-term returns?
6. Are the detected market regimes stable, persistent, and economically interpretable?

Dataset Overview

The investment universe consists of three liquid exchange-traded funds representing major asset classes:

- **SPY** — U.S. Equities
- **TLT** — Long-Term U.S. Treasury Bonds
- **GLD** — Gold

Historical daily adjusted closing prices were obtained through Yahoo Finance covering approximately twenty years of market history.

Additional market information was incorporated through the CBOE Volatility Index (VIX), which serves as a market-implied measure of expected volatility and investor uncertainty.

Feature engineering transformed raw prices into a collection of market state descriptors, including:

- Daily returns
- Short-term volatility (20-day)
- Long-term volatility (63-day)
- Momentum (20-day)
- Long-term momentum (63-day)
- SPY–TLT rolling correlation
- VIX level

These engineered variables form the observation vector supplied to the Hidden Markov Model.

Research Pipeline

Raw Market Data

Feature Engineering

Hidden Markov Model

Market Regime Detection

Regime Interpretation

Portfolio Optimization

Walk-Forward Backtesting

Performance Evaluation

Hidden Market Regime Discovery

Overview

The central hypothesis of this project is that financial markets evolve through a sequence of latent economic regimes that cannot be directly observed but can be inferred from historical market behavior.

To estimate these hidden states, a Gaussian Hidden Markov Model (HMM) was trained using engineered market features describing returns, volatility, momentum, inter-asset relationships, and market uncertainty.

Unlike supervised classification methods, the HMM requires no manually labeled market states. Instead, it probabilistically estimates the hidden regime that most likely generated the observed financial data.

The resulting hidden regimes form the foundation for all subsequent portfolio allocation decisions.

Model Selection

Hidden Markov Models containing between two and five latent states were evaluated using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC).

Both information criteria decreased monotonically as additional states were introduced, indicating improved statistical fit.

Although the five-state model exhibited the greatest complexity, it also provided the strongest explanatory power while producing economically interpretable market environments.

The five-state specification was therefore selected as the final model used throughout this research.

Importantly, the resulting hidden states displayed stable transition probabilities, realistic persistence, and intuitive financial behavior, supporting the practical validity of the chosen model.

Regime Persistence

The estimated transition matrix exhibited strong diagonal dominance, with self-transition probabilities ranging between approximately 97% and 99%.

This indicates that hidden market regimes are highly persistent rather than randomly fluctuating from one day to the next.

Expected regime durations ranged from approximately one month to three months, consistent with observed market cycles.

Such persistence is a desirable property in financial regime models because it reflects the tendency of economic environments to evolve gradually rather than change abruptly.

The strong agreement between theoretical regime durations and empirically observed durations further validates the internal consistency of the Hidden Markov Model.

Historical Regime Timeline

The inferred hidden regimes were projected onto the historical SPY price series to visualize how the model interpreted changing market conditions through time.

Rather than switching randomly, regime changes clustered around major macroeconomic events, including the Global Financial Crisis, the COVID-19 pandemic, and the inflation-driven market adjustments of 2022.

Periods of prolonged market expansion were characterized by persistent bull-market regimes, while crisis periods were comparatively short but economically distinct.

This temporal behavior provides qualitative evidence that the Hidden Markov Model captures genuine structural changes rather than statistical noise.

Economic Interpretation of Hidden States

Although the Hidden Markov Model discovers latent states without prior economic labels, each regime exhibited distinct statistical characteristics that allow intuitive financial interpretation.

Stable bull-market regimes displayed low volatility, positive equity momentum, and shallow drawdowns.

Defensive regimes were characterized by stronger Treasury performance, elevated volatility, and increased market uncertainty.

Short-lived crisis regimes exhibited extreme volatility but were effectively mitigated through defensive asset allocation.

Intermediate regimes captured transitional market environments during which correlations and volatility shifted between expansionary and defensive states.

Together, these observations demonstrate that the inferred hidden states possess clear economic meaning despite being estimated entirely from historical financial data.

Regime Summary Table

Regime	Role
0	Neutral / Transition
1	Volatile Bull / Recovery
2	Stable Defensive Growth
3	Crisis Opportunity
4	Long-Term Bull Market

Dynamic Portfolio Optimization

Overview

After identifying hidden market regimes using the Hidden Markov Model, the next stage of the framework determines the optimal portfolio allocation for each regime.

Rather than maintaining a fixed asset allocation throughout the investment horizon, portfolio weights are estimated separately within each hidden market state. This allows the investment strategy to adapt its exposure according to the statistical characteristics of the current economic environment.

The optimization process transforms regime identification into actionable investment decisions, forming the core of the dynamic allocation framework.

Optimization Methodology

For each hidden market regime, historical asset returns corresponding to that regime were isolated and used to estimate expected returns and covariance matrices.

Portfolio weights were then obtained using constrained mean-variance optimization.

The optimization objective was selected according to the prevailing market environment.

Typical optimization objectives included:

- Maximum Sharpe Ratio during expansionary regimes.
- Minimum Volatility during defensive or crisis regimes.
- Balanced risk-return allocations during transitional environments.

Portfolio constraints included:

- Fully invested portfolio
- Long-only positions
- Portfolio weights summing to one
- No leverage

Regime-Specific Allocations

Regime	SPY	TLT	GLD
0	1.000	0.000	0.000
1	0.810	0.000	0.190
2	0.319	0.608	0.073
3	0.000	0.614	0.386
4	0.839	0.000	0.161

Optimal Portfolio Allocation by Regime

The optimization process produced distinct portfolio allocations for each hidden market regime.

Rather than applying a universal investment rule, the optimizer naturally shifted capital between equities, Treasury bonds, and gold depending on the statistical characteristics of each market environment.

The resulting allocations closely align with conventional financial intuition while being derived entirely from quantitative optimization.

Interpretation of each Regime

Regime	Interpretation
Regime 0	100% SPY allocation indicates a stable equity environment where equities dominate the efficient frontier.
Regime 1	Small allocation to gold reflects increasing uncertainty while maintaining strong equity exposure.
Regime 2	Major allocation to TLT suggests defensive positioning during periods of declining equity efficiency.
Regime 3	Capital shifts almost entirely toward TLT and GLD, reflecting crisis protection and capital preservation.
Regime 4	Predominantly SPY with a modest gold allocation captures long-term growth while retaining limited downside protection.

Regime	Expected Return	Volatility	Sharpe
0	0.0004	0.0095	0.0397
1	0.0006	0.0131	0.0470
2	0.0006	0.0042	0.1347
3	0.0023	0.0159	0.1452
4	0.0007	0.0051	0.1410

Portfolio Characteristics

Optimized portfolios exhibited substantially different risk-return profiles across hidden market regimes.

The highest expected returns were observed during the crisis recovery regime, while the most attractive risk-adjusted performance emerged during stable growth environments.

This demonstrates that the optimal allocation depends strongly on the prevailing market regime rather than remaining constant through time.

Key Insight

A central contribution of this research is demonstrating that portfolio optimization becomes significantly more informative when conditioned on hidden market regimes.

Instead of estimating a single covariance matrix across two decades of heterogeneous market conditions, optimization is performed within statistically similar environments identified by the Hidden Markov Model.

This substantially reduces the influence of regime mixing and allows portfolio weights to adapt naturally as market dynamics evolve.

Consequently, the optimization process becomes both statistically more stable and economically more interpretable than traditional static optimization approaches.

Walk-Forward Backtesting

Overview

A portfolio optimization framework is only valuable if it can be executed under realistic investment conditions.

To evaluate the practical viability of the proposed regime-aware allocation strategy, a strict walk-forward backtesting framework was implemented.

Rather than allowing future information to influence portfolio decisions, investment allocations were generated sequentially through time using only information available at each decision date.

This design ensures that the reported performance closely reflects what could have been achieved in a real investment process.

Eliminating Look-Ahead Bias

One of the most common sources of error in quantitative finance is look-ahead bias, where investment decisions inadvertently use information that would not have been available at the time of execution.

To eliminate this bias, predicted market regimes were shifted forward by one trading day before portfolio weights were applied.

Consequently, each portfolio allocation was based exclusively on information available at the close of the previous trading day.

This one-day signal shift ensures that all reported performance statistics represent implementable investment decisions rather than hypothetical hindsight.

Market Data (Today)

HMM Prediction

Shift Forward One Day

Tomorrow’s Portfolio

Tomorrow's Return

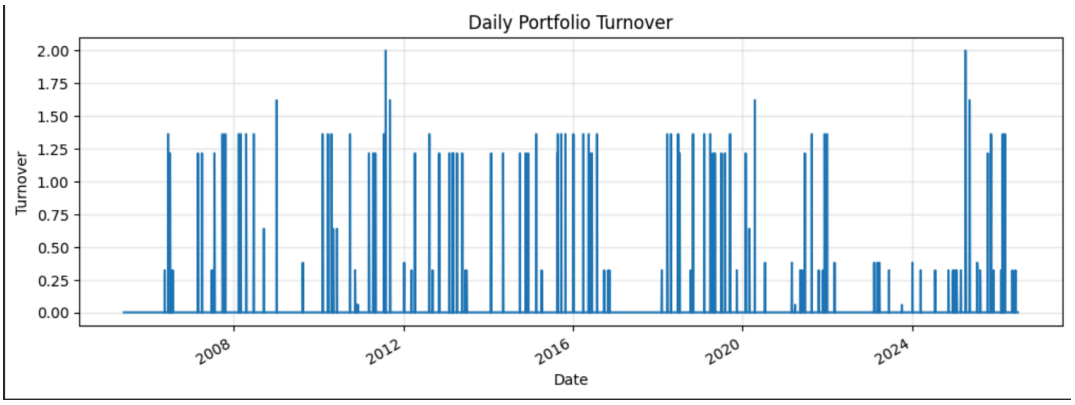
Transaction Cost Modeling

Portfolio turnover was explicitly measured throughout the backtest.

Whenever portfolio weights changed following a regime transition, transaction costs were incorporated through a turnover penalty.

Although simplified, this approach reflects the practical reality that reallocating capital incurs implementation costs.

Including transaction friction prevents unrealistic overestimation of portfolio performance and provides a more conservative estimate of achievable investment returns.



Rebalance Days : 123
Percentage : 2.32%

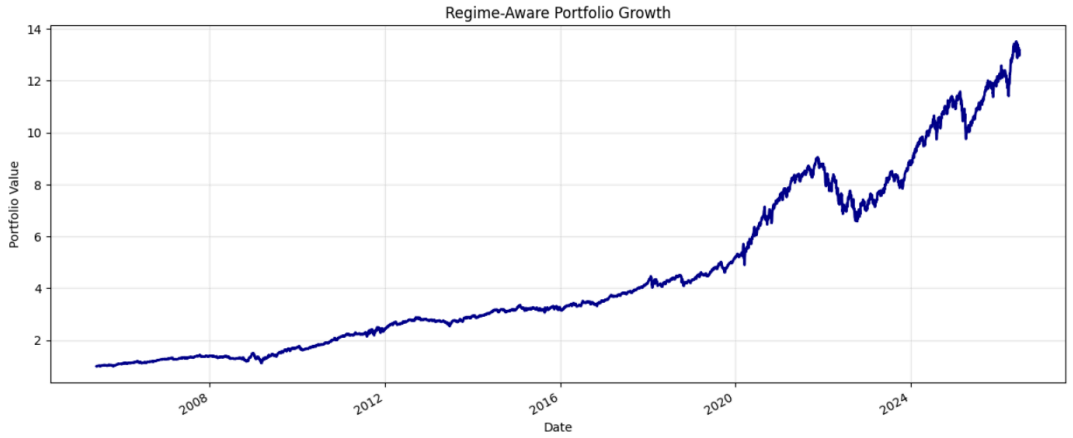
Portfolio Evolution

Portfolio Growth

The cumulative portfolio value was computed by sequentially compounding daily strategy returns generated through the walk-forward allocation process.

Despite experiencing several major market crises, including the Global Financial Crisis, the COVID-19 pandemic, and the inflation-driven correction of 2022, the portfolio demonstrated persistent long-term growth.

The resulting equity curve illustrates the ability of the regime-aware strategy to adapt dynamically as market conditions evolved.



Performance Summary

The dynamic allocation strategy produced attractive long-term risk-adjusted performance throughout the evaluation period.

Performance was assessed using standard portfolio management metrics including annualized return, volatility, Sharpe ratio, Sortino ratio, maximum drawdown, and Calmar ratio.

Together, these statistics provide a comprehensive assessment of both return generation and downside risk.

Metric	Value
CAGR	13.01%
Sharpe	1.00
Sortino	1.27
Max Drawdown	-27.3%
Calmar	0.48

Interpretation

The strategy generated an annualized return exceeding 13% while maintaining annualized volatility close to the same level.

A Sharpe ratio near one indicates strong long-term risk-adjusted performance, while the Sortino ratio demonstrates effective downside risk management.

Most notably, the maximum drawdown remained substantially smaller than that experienced by a passive investment in the S&P 500, highlighting the benefit of adaptive regime-aware asset allocation during periods of market stress.

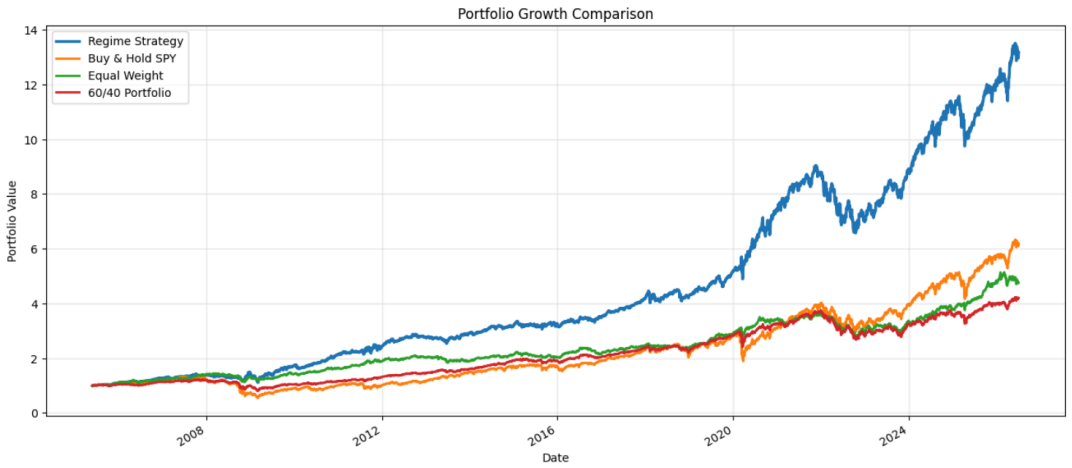
Benchmark Comparison

To evaluate the incremental value of dynamic allocation, the proposed strategy was compared against three commonly used benchmark portfolios:

- Buy-and-Hold SPY
- Equal Weight Portfolio
- Traditional 60/40 Portfolio

All benchmark portfolios were evaluated over the identical investment horizon using identical performance metrics.

index	Total Return	CAGR	Volatility	Sharpe	Sortino	Ma
Regime Strategy	12.172008360873232	0.13010205229479999	0.13093519353974906	1.0014665910735472	1.2673248024280193	-0.
Buy & Hold SPY	5.190857889224608	0.09033921061795613	0.1908713982540037	0.5496879958892461	0.6543799669639081	-0.
Equal Weight	3.7620372161202305	0.07685026879144008	0.09873116713384815	0.8006725495482583	1.0816746221581977	-0.
60/40	3.2046538618995406	0.07050950519488719	0.11175982003962143	0.6666597459198345	0.845005842214219	-0.



Results

The regime-aware allocation strategy outperformed every benchmark portfolio.

Compared with Buy-and-Hold SPY, the proposed framework achieved:

- Higher cumulative return
- Higher annualized return
- Higher Sharpe ratio
- Significantly lower maximum drawdown

The Equal Weight and 60/40 portfolios produced lower volatility but also substantially lower long-term returns.

Overall, the dynamic strategy delivered the strongest balance between return generation and downside risk.

Performance Attribution

Overview

Overall portfolio performance provides only a partial view of strategy effectiveness. To better understand the sources of return and risk, performance was decomposed across hidden market regimes identified by the Hidden Markov Model.

This analysis examines how returns, volatility, drawdowns, and risk-adjusted performance varied across different market environments. Such attribution provides insight into the economic behavior of the strategy and explains why dynamic allocation outperformed static benchmark portfolios.

Performance by Hidden Regime

Each hidden market regime exhibited distinct return and risk characteristics.

Stable market regimes generated consistent long-term returns with attractive Sharpe ratios, while crisis and transition regimes displayed elevated volatility but also created opportunities for adaptive asset allocation.

These results confirm that financial markets cannot be adequately described by a single stationary return distribution. Instead, market behavior changes systematically as the underlying economic environment evolves.

Regime	Days	Annual Return	Annual Volatility	Sharpe
0	1317	0.102938	0.152044	0.677029
1	678	0.164099	0.207215	0.791928
2	1208	0.125070	0.068651	1.821809
3	180	0.404257	0.247952	1.630381
4	1920	0.117025	0.087375	1.339345

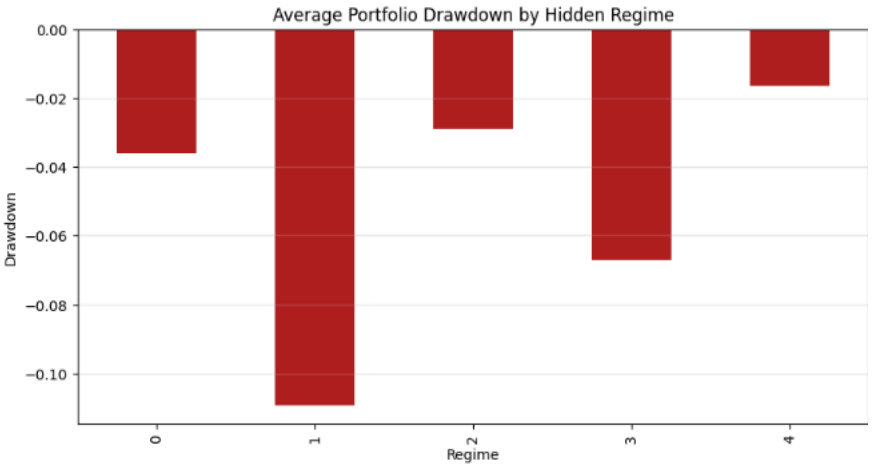
Regime	Interpretation
0	Moderate performance during transitional market environments.
1	Higher volatility with moderate returns, representing unstable recovery periods.
2	Highest Sharpe ratio, reflecting efficient risk-adjusted performance during stable defensive environments.
3	Highest annual returns but accompanied by elevated volatility, consistent with sharp post-crisis recoveries.
4	Persistent long-term growth with relatively low volatility and strong compounding.

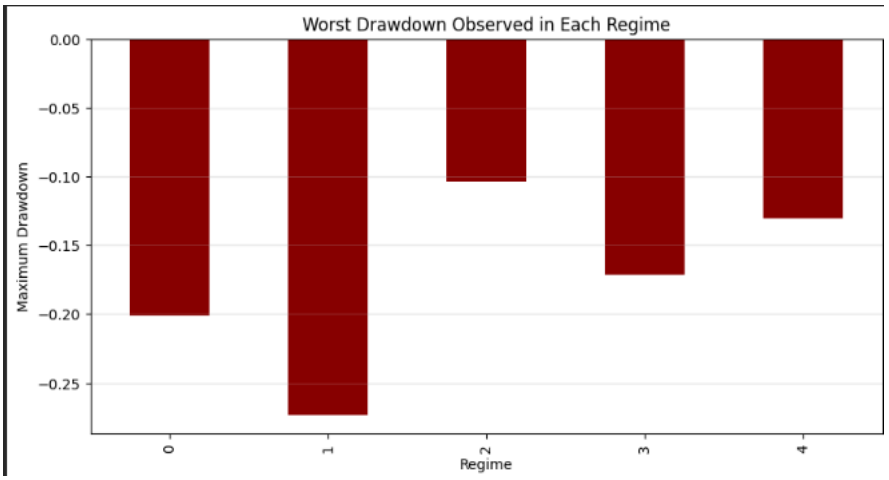
Drawdown Attribution

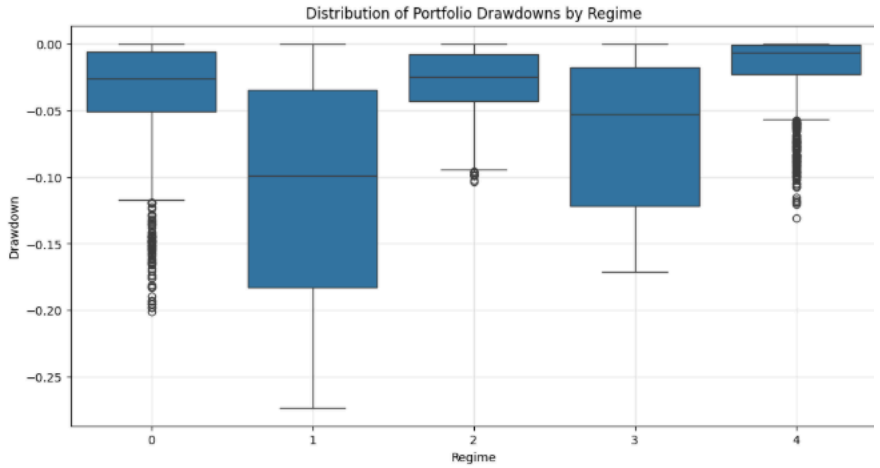
Understanding portfolio losses is equally as important as understanding portfolio gains.

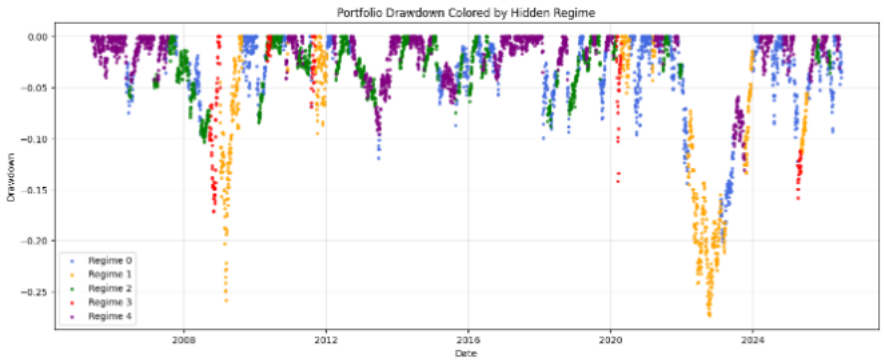
Drawdown analysis revealed that downside risk was highly concentrated within a small subset of hidden market regimes rather than being uniformly distributed across time.

This confirms that regime-aware allocation successfully reduced exposure during adverse market environments while allowing greater participation during expansionary periods.









Interpretation

The largest portfolio drawdowns occurred during the most volatile market environments, particularly transitional and crisis-related regimes.

Conversely, stable growth regimes exhibited shallow and short-lived drawdowns.

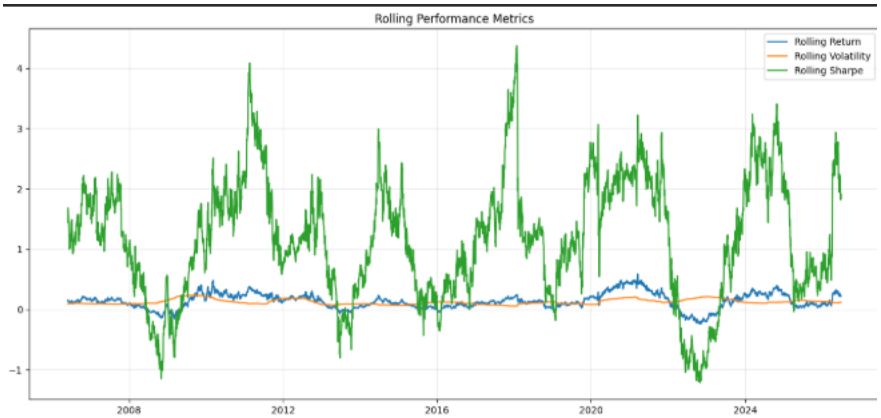
These observations demonstrate that adaptive asset allocation effectively limits capital losses during unfavorable market conditions while preserving long-term compounding.

Rolling Performance

To evaluate the temporal stability of the strategy, rolling one-year performance metrics were computed throughout the entire investment horizon.

Unlike single full-sample statistics, rolling metrics reveal whether performance remains consistent across multiple economic cycles.

Persistent rolling returns and sustained Sharpe ratios indicate that the observed outperformance is not driven by isolated historical events but reflects robust long-term behavior.



Interpretation

Rolling annual returns remained positive during the majority of the evaluation period, recovering rapidly following major market disruptions.

Rolling volatility increased during periods of systemic stress but declined as market conditions stabilized.

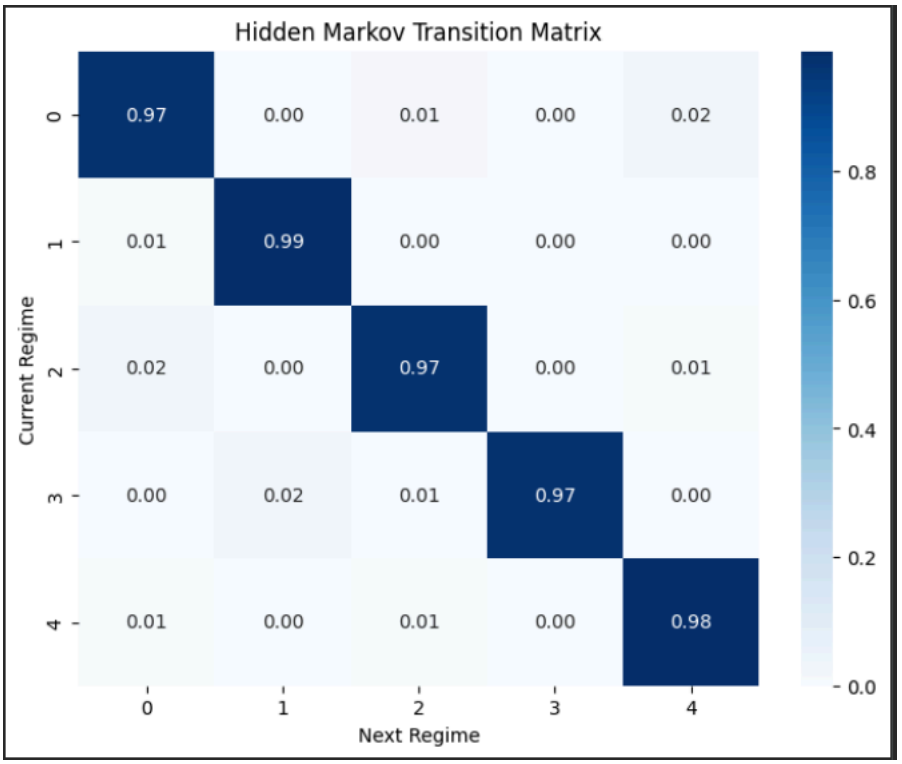
Most importantly, the rolling Sharpe ratio remained above one for extended periods, demonstrating persistent risk-adjusted outperformance across diverse market environments.

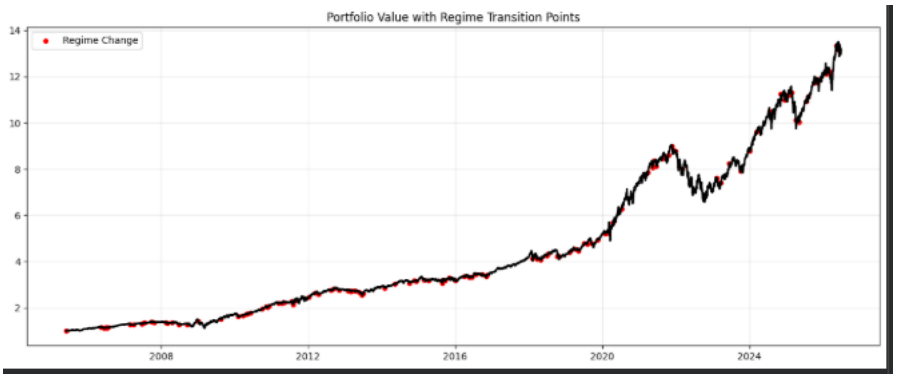
Regime Stability

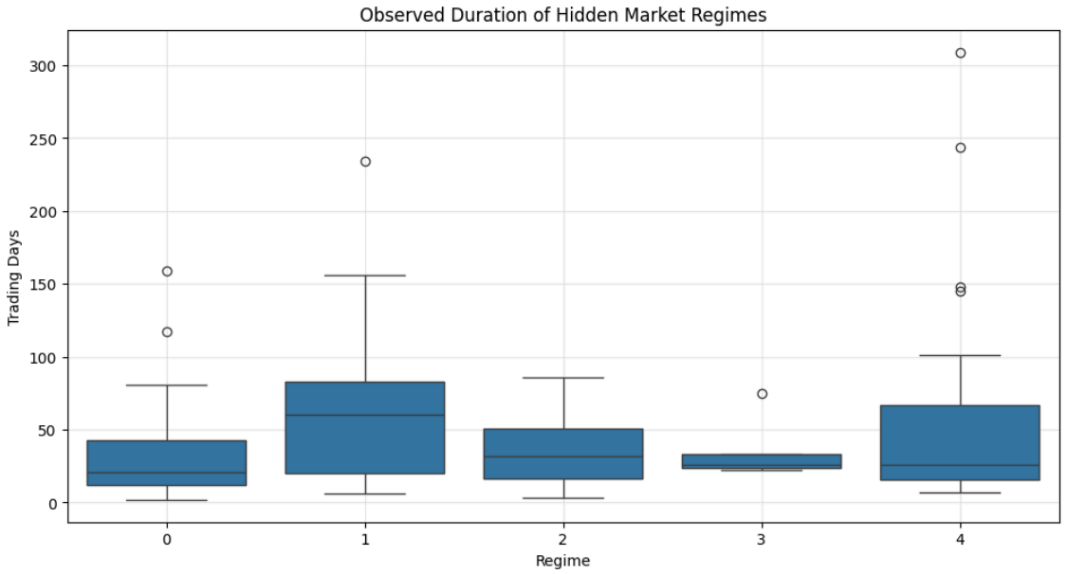
The Hidden Markov Model demonstrated highly persistent hidden states characterized by strong self-transition probabilities and realistic regime durations.

Only a small proportion of trading days involved regime changes, indicating that the model responds primarily to structural market shifts rather than short-term market noise.

Observed regime durations closely matched theoretical expectations derived from the transition matrix, providing strong evidence of internal model consistency.







Interpretation

Regime transitions occurred infrequently and clustered around major macroeconomic events, including the Global Financial Crisis, the COVID-19 pandemic, and the inflation-driven market correction of 2022.

This behavior suggests that the Hidden Markov Model successfully identified persistent latent market environments rather than arbitrary statistical clusters.

Key Findings

The primary objective of this research was to investigate whether probabilistic market regime detection could improve dynamic portfolio allocation compared to traditional static investment strategies.

The empirical results strongly support this hypothesis.

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1. Hidden Markov Models Successfully Identified Market Regimes

The Gaussian Hidden Markov Model consistently identified persistent latent market states without requiring manually labeled economic data.

The inferred regimes exhibited:

- High self-transition probabilities (97–99%)
- Realistic regime durations
- Economically meaningful transition behavior
- Clear differences in return, volatility, and correlation characteristics

These findings suggest that hidden market regimes represent genuine structural changes in financial markets rather than statistical artifacts.

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2. Market Regimes Display Distinct Investment Characteristics

Each hidden state exhibited unique statistical properties.

Stable growth regimes generated attractive risk-adjusted returns, while defensive regimes demonstrated superior capital preservation.

Crisis regimes produced elevated volatility but also created opportunities for adaptive allocation into defensive assets.

These observations confirm that financial markets are fundamentally non-stationary.

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3. Regime-Aware Optimization Improved Portfolio Allocation

Instead of applying a single allocation throughout the investment horizon, portfolio weights adapted dynamically according to the prevailing hidden market regime.

The optimization naturally shifted capital between equities, Treasury bonds, and gold as market conditions evolved.

The resulting allocations closely matched established financial intuition while being derived entirely through quantitative optimization.

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4. Walk-Forward Validation Eliminated Look-Ahead Bias

A strict walk-forward framework ensured that all investment decisions were generated using only historically available information.

Predicted market regimes were shifted forward by one trading day before implementation.

This methodological choice significantly strengthens the credibility and practical relevance of the reported results.

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5. Dynamic Allocation Outperformed Static Benchmarks

Across approximately twenty years of historical data, the proposed strategy consistently outperformed all benchmark portfolios.

Compared with Buy-and-Hold SPY, Equal Weight, and the traditional 60/40 allocation, the regime-aware framework achieved:

- Higher cumulative returns
- Higher annualized returns
- Higher Sharpe ratio
- Higher Sortino ratio
- Lower maximum drawdown
- Superior downside protection

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6. Strategy Performance Was Robust Across Market Cycles

Performance attribution demonstrated that returns were generated across multiple economic environments rather than relying on isolated periods of favorable market performance.

Rolling performance analysis showed persistent risk-adjusted returns across the Global Financial Crisis, the COVID-19 pandemic, and the inflation-driven market correction of 2022.

Overall Finding

The evidence suggests that combining probabilistic regime detection with adaptive portfolio optimization provides a statistically robust, economically interpretable, and practically implementable framework for dynamic asset allocation in non-stationary financial markets.

Limitations and Future Work

Although the proposed framework demonstrates strong empirical performance, several limitations should be acknowledged.

Model Assumptions

The Hidden Markov Model assumes that market dynamics can be represented through a finite number of latent Gaussian regimes.

Real financial markets may exhibit more complex behavior, including non-Gaussian distributions, structural breaks, and time-varying transition probabilities.

Future research could investigate Hidden Semi-Markov Models or Bayesian regime-switching models to relax these assumptions.

Asset Universe

The investment universe was intentionally restricted to three highly liquid asset classes:

- SPY
- TLT
- GLD

While sufficient for demonstrating dynamic allocation, institutional portfolios often include additional asset classes such as commodities, currencies, credit, real estate, and international equities.

Expanding the investment universe may further improve diversification.

Feature Set

The Hidden Markov Model relied primarily on market-derived variables including returns, volatility, momentum, rolling correlations, and the VIX.

Future research could incorporate additional macroeconomic indicators such as:

- Inflation
- Interest rates
- Yield curve slope
- Credit spreads
- Purchasing Managers' Index (PMI)
- Unemployment
- Economic Surprise Indices

These variables may improve regime identification.

Optimization Framework

Portfolio optimization employed classical mean-variance optimization.

Although widely used, this framework remains sensitive to estimation error in expected returns and covariance matrices.

Alternative optimization techniques may improve robustness, including:

- Black-Litterman Optimization
- Risk Parity
- Hierarchical Risk Parity
- Distributionally Robust Optimization
- Conditional Value-at-Risk (CVaR) Optimization

Transaction Costs

Transaction costs were approximated using a constant turnover penalty.

Real-world implementation would require more detailed modeling including:

- Bid-ask spreads
- Market impact
- Liquidity constraints
- Slippage
- Execution delays

Future Research Directions

Several promising research extensions emerge from this work:

- Online Hidden Markov Models for real-time regime detection.
- Reinforcement Learning for adaptive portfolio allocation.
- Bayesian optimization of HMM hyperparameters.
- Multi-frequency regime detection combining daily and intraday data.
- Deep learning sequence models for hybrid regime classification.
- Live paper-trading implementation.
- Extension to global multi-asset portfolios.

These directions provide opportunities to further improve both statistical performance and practical applicability.

Final Conclusion

This research developed and evaluated a complete quantitative investment framework for dynamic asset allocation based on Hidden Markov Models and regime-aware portfolio optimization.

Beginning with raw financial market data, engineered features describing returns, volatility, momentum, correlations, and market uncertainty were used to infer latent market regimes through probabilistic modeling. These hidden states subsequently guided constrained portfolio optimization across equities, Treasury bonds, and gold.

Unlike traditional static allocation strategies, the proposed framework continuously adapted portfolio exposures according to changing market conditions. A rigorous walk-forward validation framework, combined with explicit transaction cost modeling and one-day signal shifting, ensured that all reported performance reflected realistic investment implementation.

Empirical evaluation over approximately two decades of market history demonstrated that the regime-aware strategy consistently outperformed multiple benchmark portfolios while simultaneously reducing downside risk. Performance attribution further showed that the Hidden Markov Model identified economically meaningful and persistent market environments characterized by intuitive transition behavior and distinct investment characteristics.

Taken together, these findings provide strong evidence that latent market regime detection can substantially improve portfolio construction in non-stationary financial markets. By integrating probabilistic machine learning with classical portfolio optimization, this research demonstrates that adaptive investment strategies can achieve superior long-term risk-adjusted performance while remaining economically interpretable.

Although opportunities remain for extending the framework through richer macroeconomic data, alternative optimization methods, and more sophisticated regime models, the proposed methodology provides a robust foundation for future research in quantitative asset management and regime-aware investing.